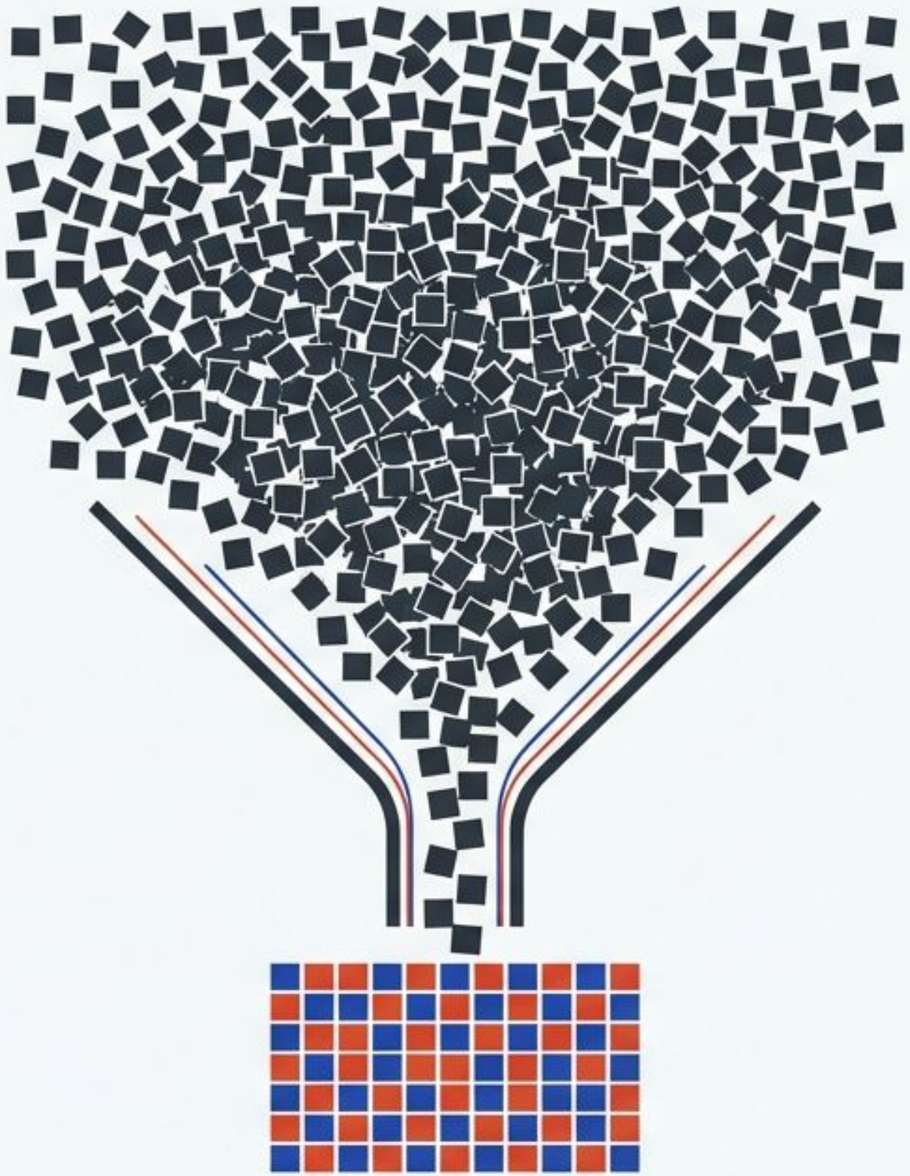
Decoding Data Compression

Unit 5: Multimedia Systems
Core Mechanics of Storage and Transmission

From Theoretical Entropy to Modern Standards



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Surviving the Multimedia Data Deluge

Storage Space

Makes massive media libraries feasible on limited devices.

Bandwidth Efficiency

Enables high-speed transfers and seamless streaming.

Cost Savings

Lowers operational hardware and network costs.

Backup & Recovery

Accelerates data safety protocols and reduces physical disk needs.

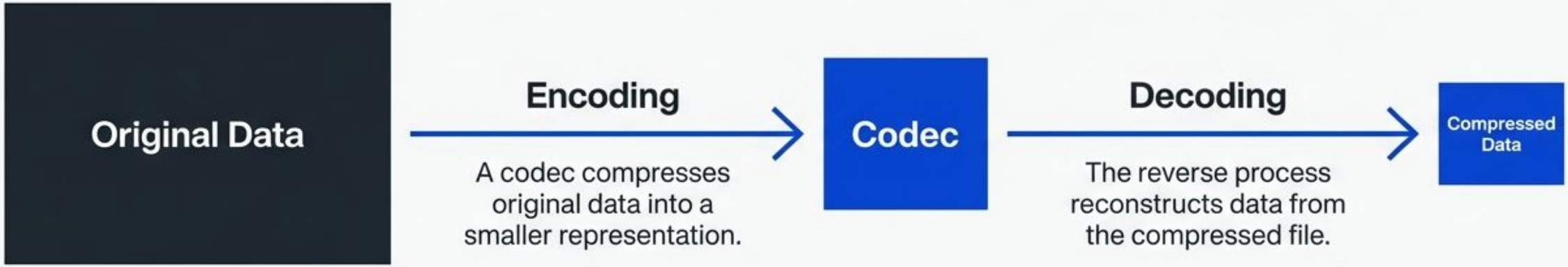
Real-World Context

Without compression, 1 minute of uncompressed HD video equals several gigabytes. Compression makes Blu-ray and Netflix 4K streaming mathematically possible.

Eliminating Redundancy to Shrink Data

The Core Concept

Finding and eliminating predictable or repeated information.



10:1

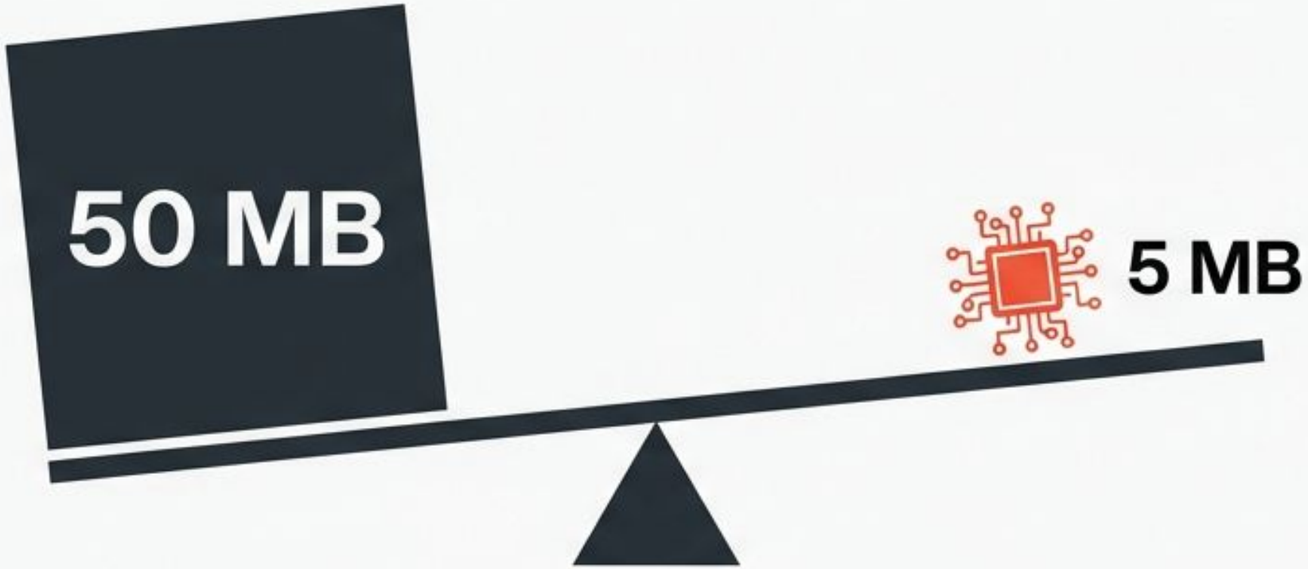
Compression Ratio

Original Size ÷ Compressed Size

Balancing File Size Against Processing Power

Uncompressed Impact: 50 MB

A 5-minute CD-quality song.



Compressed Impact: 5 MB

The same song as a 128 kbps MP3 (90% reduction).

The Trade-off

Smaller storage requires complex algorithms.

The Cost

High compression increases CPU usage and processing time.

Real-World Context

A 128GB smartphone holds thousands of photos because the massive processing power of the phone's chip instantly compresses them into small JPEGs.

Four Rules for Effective Compression Codes

Uniquely Decodable

Completely unambiguous; one compressed bitstream equals exactly one original message.

Instantaneous

Decodes bits immediately without looking ahead (e.g., Prefix-free Huffman codes).

Efficient

Computationally viable to encode and decode on target hardware.

Adaptable

Capable of adjusting to changing statistical patterns in the data flow.

The Great Divide Between Perfect and Imperfect

Lossless

- Perfectly preserves original data (Fully reversible).
- Ratios: Low to moderate (2:1 to 4:1).
- Uses: Text, Code, ZIP files.

A zipped Word doc unzips identically.

Lossy

- Permanently discards “unimportant” data (Irreversible).
- Ratios: High to extreme (10:1 to 100:1+).
- Uses: Photos, Video, Audio.

A Netflix video is an irreversible approximation.

Combining Techniques for Hybrid Coding

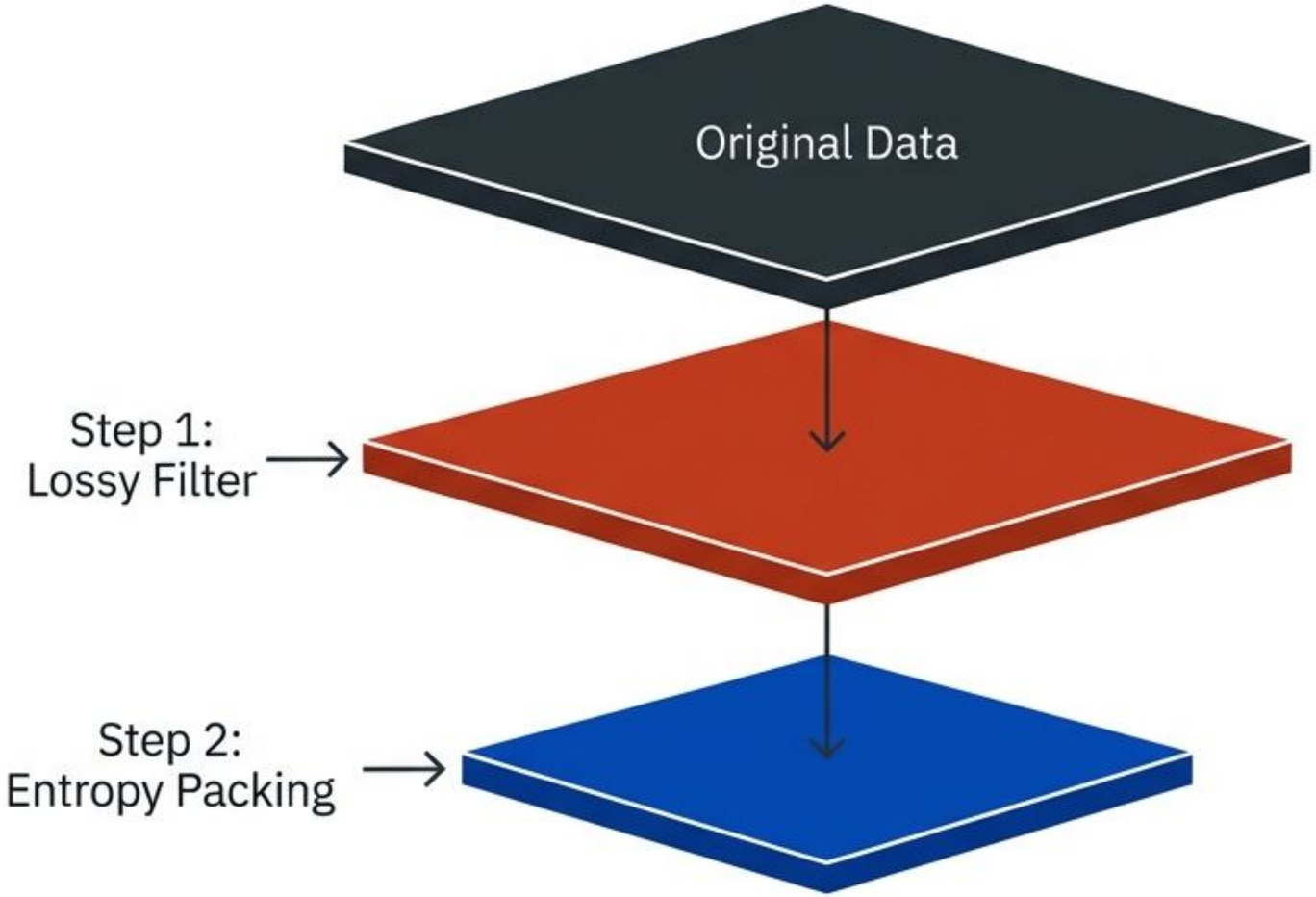
Fuses multiple techniques to achieve maximum compression.

Step 1: Lossy Filter

Removes perceptually irrelevant details to slash file size.

Step 2: Entropy Packing

Uses lossless coding to perfectly pack the remaining data.



Transforming Pixels into Frequencies with DCT

Block Preparation

The image is sliced into precise 8x8 pixel grids.

The DCT Formula

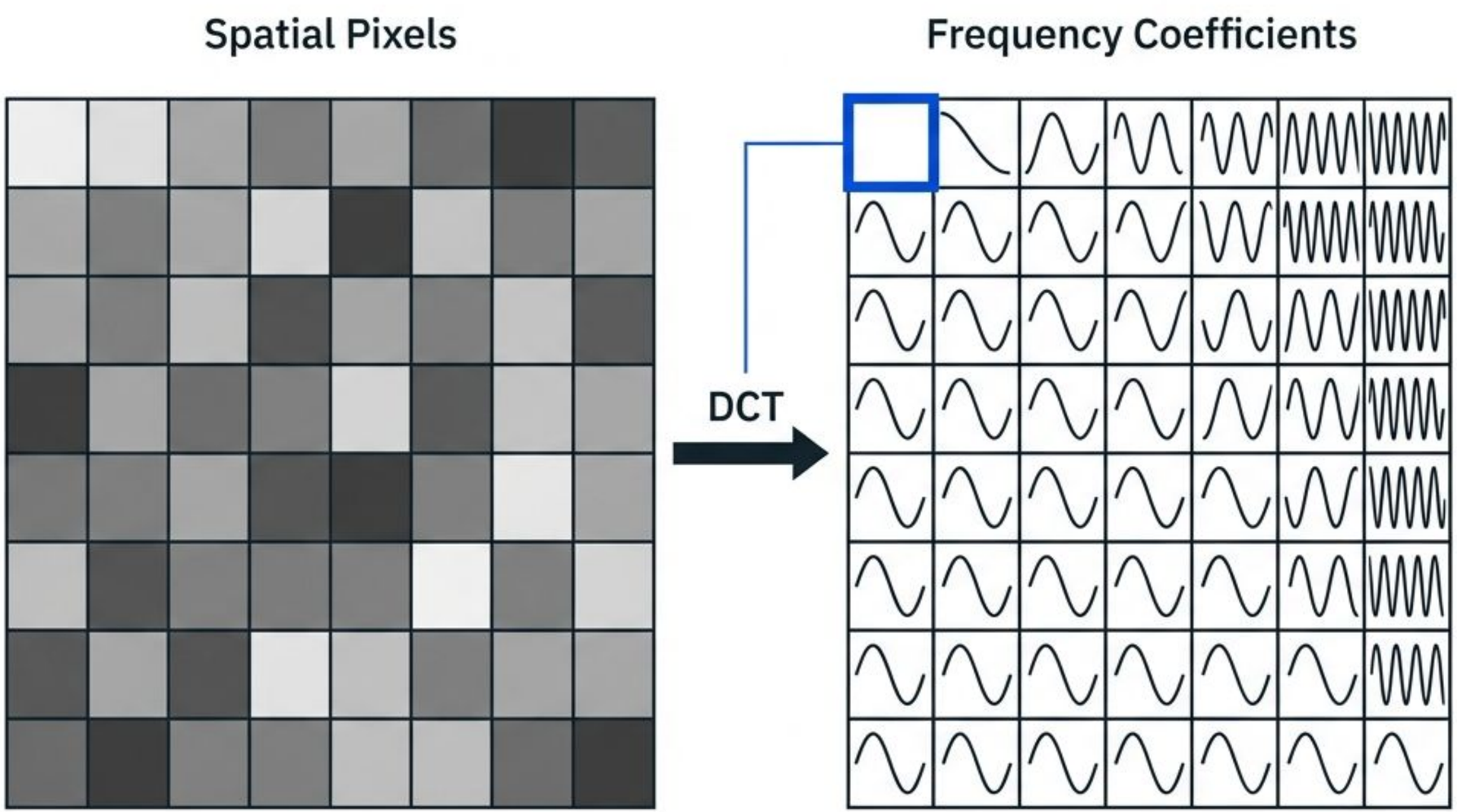
Converts standard spatial pixels into frequency waves.

DC Coefficient

The top-left value representing the block's average color.

AC Coefficients

The remaining 63 values representing fine details.



Discarding Data Through Quantization and Scanning

Techniques to reduce file size by permanently discarding less significant details.

Quantization (The Lossy Step)

High-frequency details are divided and become zero; data is permanently discarded.

Zig-Zag Scan

Reads the 8x8 block diagonally to group all new zeros together.

Run-Length Encoding (RLE)

Efficiently counts the long runs of discarded zero data.

Entropy Coding (The Lossless Step)

Variable-length codes finalize the compressed file.

92	56	33	15	27	18	10	07
56	48	33	21	17	12	0	0
33	37	27	18	16	0	0	0
15	22	10	07	0	0	0	0
15	18	23	0	0	0	0	0
18	18	0	0	0	0	0	0
10	07	0	0	0	0	0	0
07	0	0	0	0	0	0	0

Quantization & Zig-Zag Scan of 8x8 Block

Enhancing User Experience on Slow Networks

Sequential DCT: Encodes and decodes the image in a single, top-to-bottom pass.

Progressive DCT: Encodes the image in multiple, hierarchical layered scans.

The Load Experience: Displays a low-quality rough version first, refining details sequentially.

The Trade-off: Provides instant feedback on slow connections, but requires more processing power.

Sequential

Sequential

Progressive

Progressive

The Standard for Continuous-Tone Photographs

JPEG Protocol

Joint Photographic Experts Group standard.

Primary Use

Optimized for continuous-tone still images.

Advantages

Massive 10:1 to 20:1 compression ratios with universal browser support.

User Control

Allows explicit trade-offs between file size and image quality.

Limitations

No transparency support; generates blocky artifacts at extreme compression.

Exploiting Spatial and Temporal Redundancy

MPEG Protocol

Moving Picture Experts Group standard for video.

I-Frames (Intra-coded)

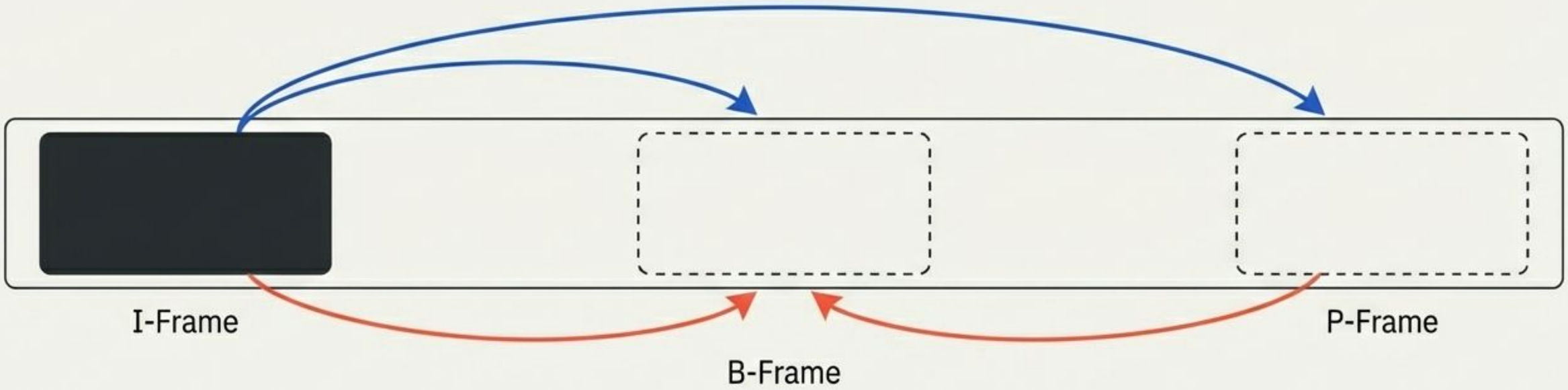
Standalone spatial frames containing complete image data, like a JPEG.

P-Frames (Predicted)

Temporal compression storing only the differences from past frames.

B-Frames (Bi-directional)

Analyzes past and future frames for absolute maximum compression.



Tracking Motion to Achieve Massive Compression

Motion Compensation

The codec actively tracks moving pixel blocks across frames.

Data Savings

Encoding a vector path is vastly smaller than redrawing the pixels.

H.264/AVC (MPEG-4 Part 10)

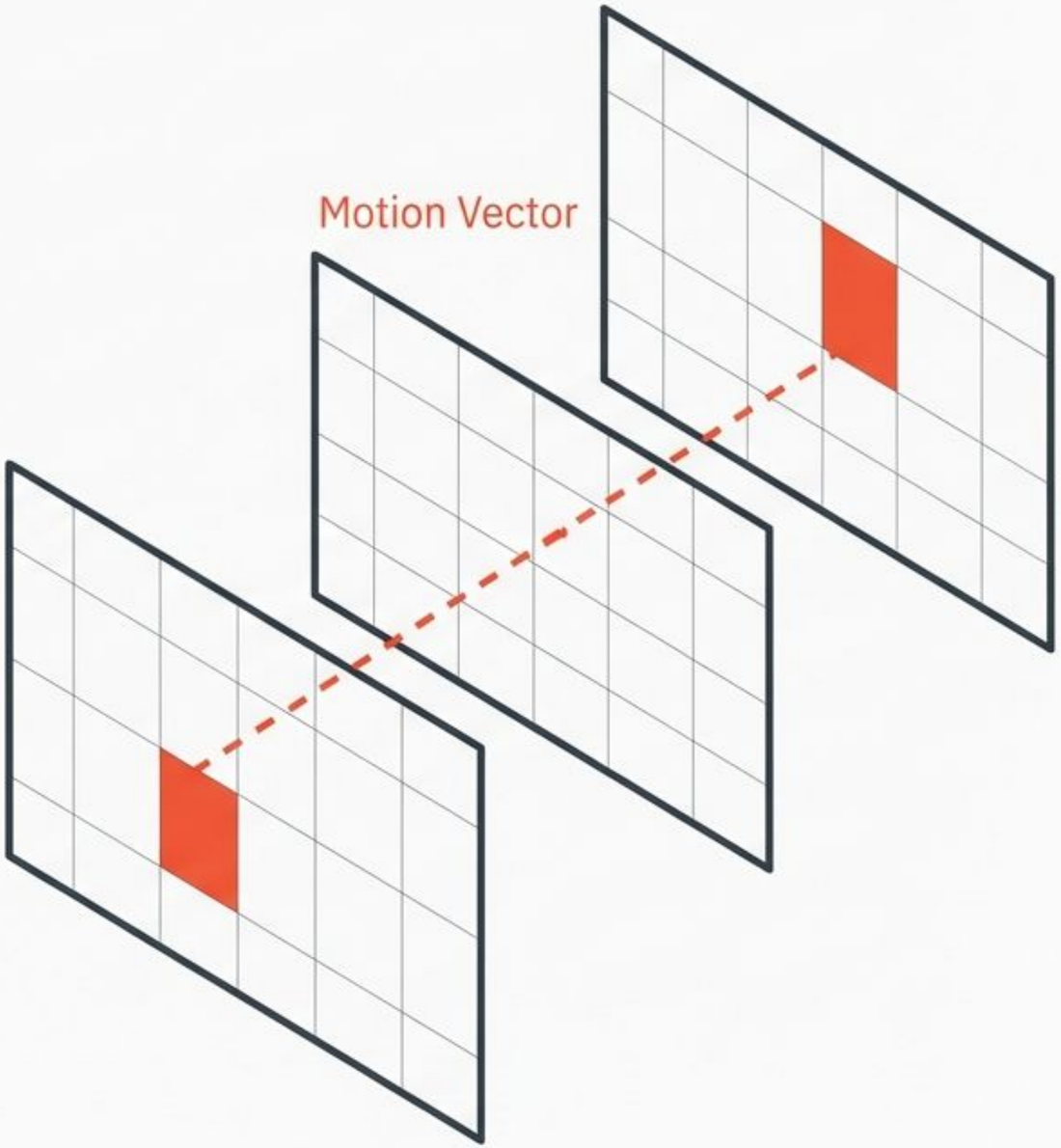
The dominant modern standard for high-definition streaming.

High Complexity, High Reward

Requires immense CPU power but enables seamless video delivery.

Real-World Ecosystem

H.264 is the precise mathematical codec powering your Netflix streams, Blu-ray discs, and Zoom video calls.



Unit 5 Exam Blueprint and Core Takeaways

1. The Goal

Eliminate spatial and temporal redundancy to reduce storage and bandwidth.

2. The Metric

Compression Ratio dictates the efficiency and success of a codec.

3. The Divide

Lossless (perfect mathematical reversal) vs. Lossy (perceptual data discarding).

4. The Hybrid Engine

Modern codecs pair Lossy data removal with Lossless Entropy packing.

5. JPEG Architecture

8x8 Blocks → DCT Frequencies → Quantization → Zig-Zag Entropy.

6. MPEG Architecture

Spatial I-Frames paired with Temporal P/B-Frames and motion vectors.